RFID-Based

2D Hand-Trajectory Tracking

on a Tag-Instrumented Desktop Surface

A time-aware, confidence-gated dual-branch deep-learning pipeline that reconstructs hand motion from 8 passive RFID tags observed by 2 antennas, using a camera-ArUco ground truth.

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Why track a hand with RFID?

Cameras, IMUs and specialized gloves dominate hand tracking — but every one of them demands line-of-sight, wearables, or power-hungry hardware.

Passive RFID tags are the opposite extreme: battery-free, ~$0.10 each, invisible under a surface, and readable through occlusion. If we can recover where the hand is from the RF signals alone, we unlock a low-cost, unobtrusive interaction layer for any flat surface — desks, tabletops, shelves.

The catch: the signal is brutal. Phase wraps every 2048 units, RSSI drifts with multipath, and a tag reply is irregular — we may get no fresh read for seconds. A naive model will happily collapse to the mean.

Irregular sampling

Reads arrive asynchronously; dt varies. Per-tag fill rate ranges from < 1% to ~99% across channels.

Phase is ambiguous

Raw phase is a sawtooth mod 2048. Small motions look like giant jumps without wrap handling.

Dead & ghost channels

Some tag/antenna pairs never report. Others echo stale data — training must know the difference.

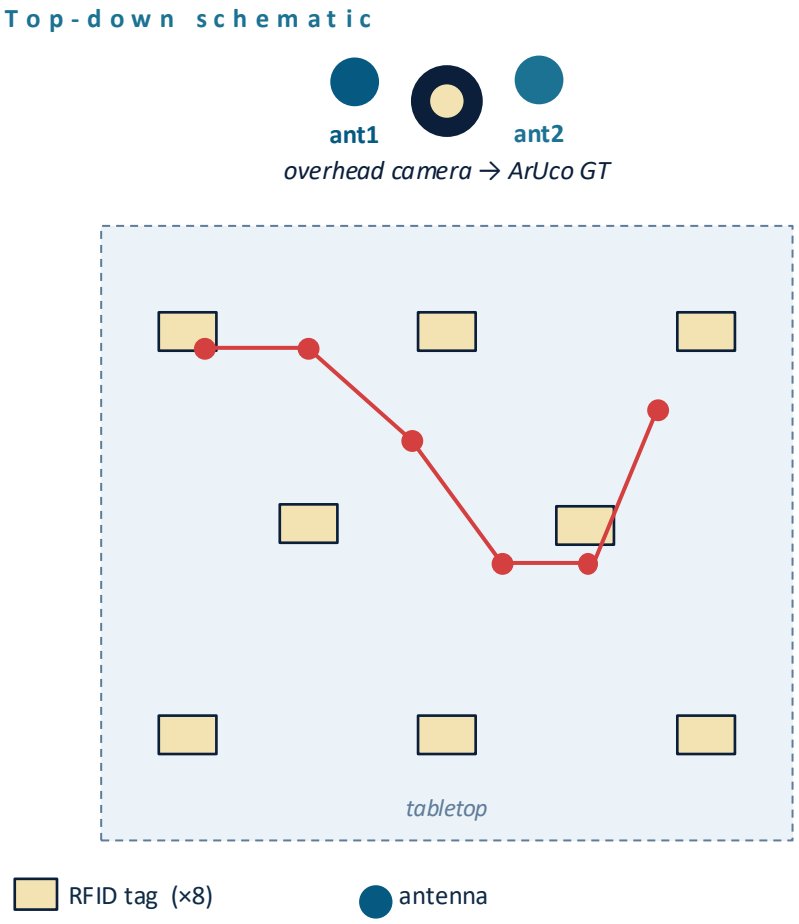
8 tags, 2 antennas, one ArUco ground truth

Hardware layout

- 8 passive UHF RFID tags adhered to a desktop in a fixed grid pattern.
- 2 reader antennas (ant1, ant4) positioned above/alongside the surface; both query the same 8-tag population.
- Per read: RSSI (dB) and phase (raw integer, 2048-period) at host timestamp.
- Overhead camera + ArUco markers on the wrist stream WRIST_x / WRIST_y in table-plane coordinates — the regression target.

Dataset collected

- 2 sessions · 435 synchronized samples · ~125 s per session of free-form hand motion on the tabletop.
- 16 antenna–tag channel pairs total (2 × 8); 1 pair excluded upstream due to consistently malformed phase.
- Mean RSSI-fill rate per channel ≈ 72% — confirming that missing reads are the rule, not the exception.



From raw reads to 14-feature channel tensors

16

antenna × tag channels fed to the model

14

features per channel per timestep

2048

period of the raw phase integer

Raw & masked

RSSI_z (z-scored per ch.)

sin(ϕ), cos(ϕ) on 2π·ϕ/2048

m_rssi, m_phase real-read mask

Imputation

imp_rssi, imp_phase flags

forward-fill last valid read

exposes 'stale' vs 'fresh'

Time awareness

tslv time-since-last-valid

c = exp(−λ·tslv) confidence

channel death ⇒ attn mask

Dynamics

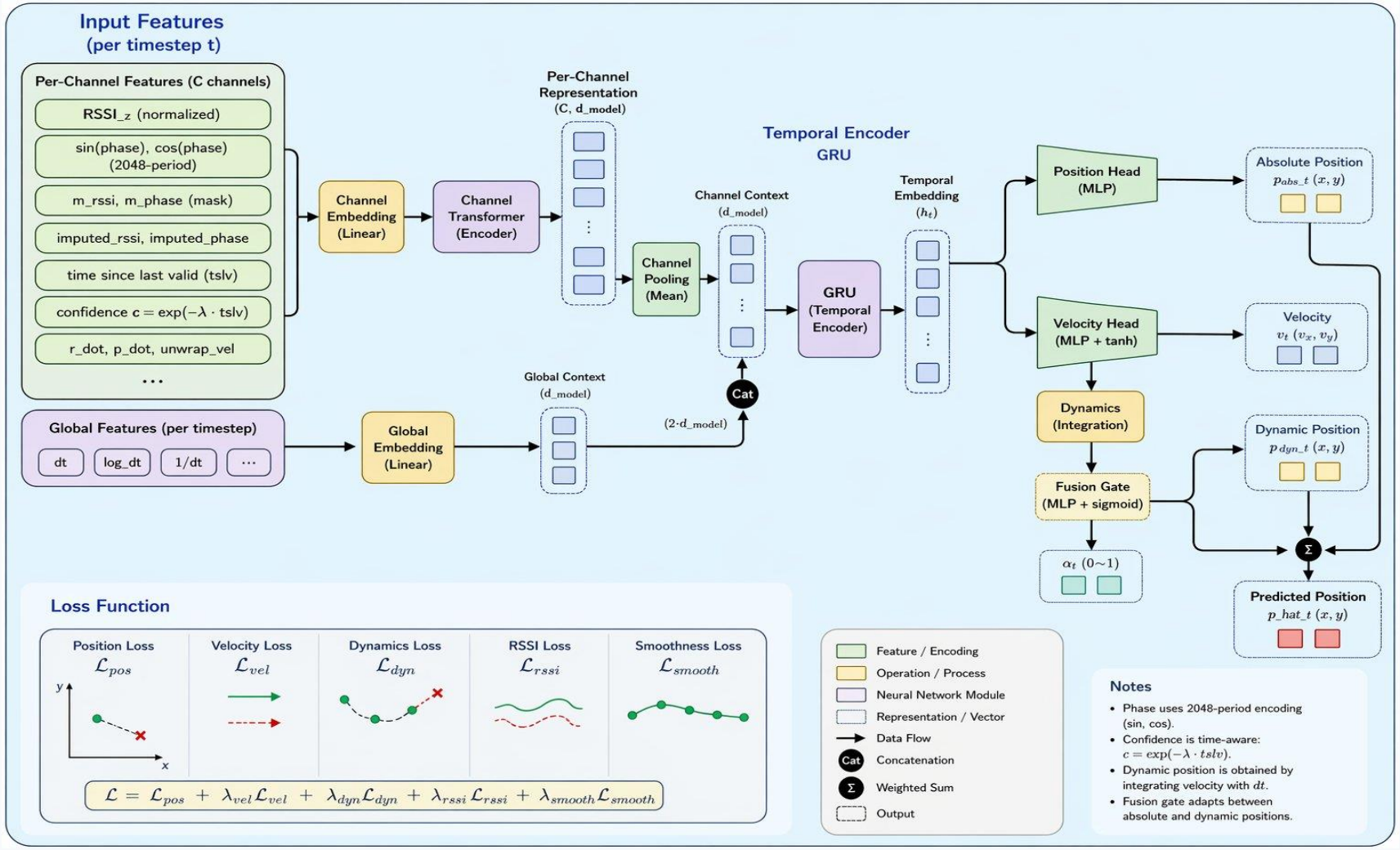
r_dot ΔRSSI / Δt

p_dot wrapped Δϕ / Δt

unwrap_vel unwrapped ϕ-vel

Windows of T = 16 consecutive samples; only kept if WRIST(x,y) is valid on every step and at least one live channel is present.

Time-aware, confidence-gated, dual-branch tracker



Channel Transformer

Lets the 16 antenna–tag channels attend to each other per timestep; dead channels are masked out of attention.

Temporal GRU

Fuses the pooled channel context with global dt features → hidden state h_t carrying short-term motion memory.

Dual-branch fusion

α·p_abs + (1−α)·p_dyn. α is gated by confidence & real-read rate — low evidence ⇒ trust physics integration of v.

Four losses: position, velocity, dynamics, smoothness

$\mathcal{L}$_pos

$$\mathcal{L}_{\text{pos}} = \text{SmoothL1}(\hat{p},\, xy) \;+\; 0.15 \cdot \text{MSE}(\hat{p},\, xy)$$

The anchor. Penalizes every predicted point against the ArUco target.

$\mathcal{L}$_vel

$$\mathcal{L}_{\text{vel}} = \text{MSE}\Big(\hat{v}_t,\; \frac{xy_t - xy_{t-1}}{\Delta t}\Big)$$

Forces the velocity head to match the numerical derivative of ground truth.

$\mathcal{L}$_dyn

$$\mathcal{L}_{\text{dyn}} = \text{MSE}(\hat{p}_t \;-\; (\text{sg}(\hat{p}_{t-1}) + \hat{v}_{t-1}\,\Delta t))$$

Couples the two branches: integrated velocity must match emitted position.

$\mathcal{L}$_smooth

$$\mathcal{L}_{\text{smooth}} = \text{MSE}(\hat{v}_t \;-\; \hat{v}_{t-1})$$

Damps high-frequency jitter in the velocity head.

Combined objective

$$\mathcal{L} = \mathcal{L}_{\text{pos}} + 0.40\,\mathcal{L}_{\text{vel}} + 0.25\,\mathcal{L}_{\text{dyn}} + 0.08\,\mathcal{L}_{\text{smooth}}$$

Position dominates; the other three act as physics and smoothness regularizers.

Training setup

optimizer	AdamW, lr = 1e-3, weight-decay = 1e-4
schedule	60 epochs · batch 16 · grad-clip 2.0
window T	16 samples (~1.6–3 s of hand motion)
split policy	temporal — 168 train / 42 val windows
channels used	15 antenna-tag pairs (1 dropped upstream)
model	Tx d_model = 32, 4 heads · GRU 128

Temporal-split evaluation — the honest metrics

0.131
pos. RMSE L2
(units = normalized tabletop)

0.108
pos. MAE
mean L2 per point

0.036
vel. RMSE L2
vx, vy combined

2.68 %
hit-rate @ ε
ε = 0.02 (≈ tag spacing / 25)

Full temporal-split report

42 val windows · 672 points · window T = 16

metric	x	y	L2 / combined
position RMSE	0.1096	0.0721	**0.1312**
position MAE	0.0804	0.0605	**0.1081**
velocity RMSE	0.0279	0.0230	**0.0361**
per-window mean L2 (μ ± σ)	—	—	**0.108 ± 0.033**
final-step L2 (μ ± σ)	—	—	**0.144 ± 0.092**

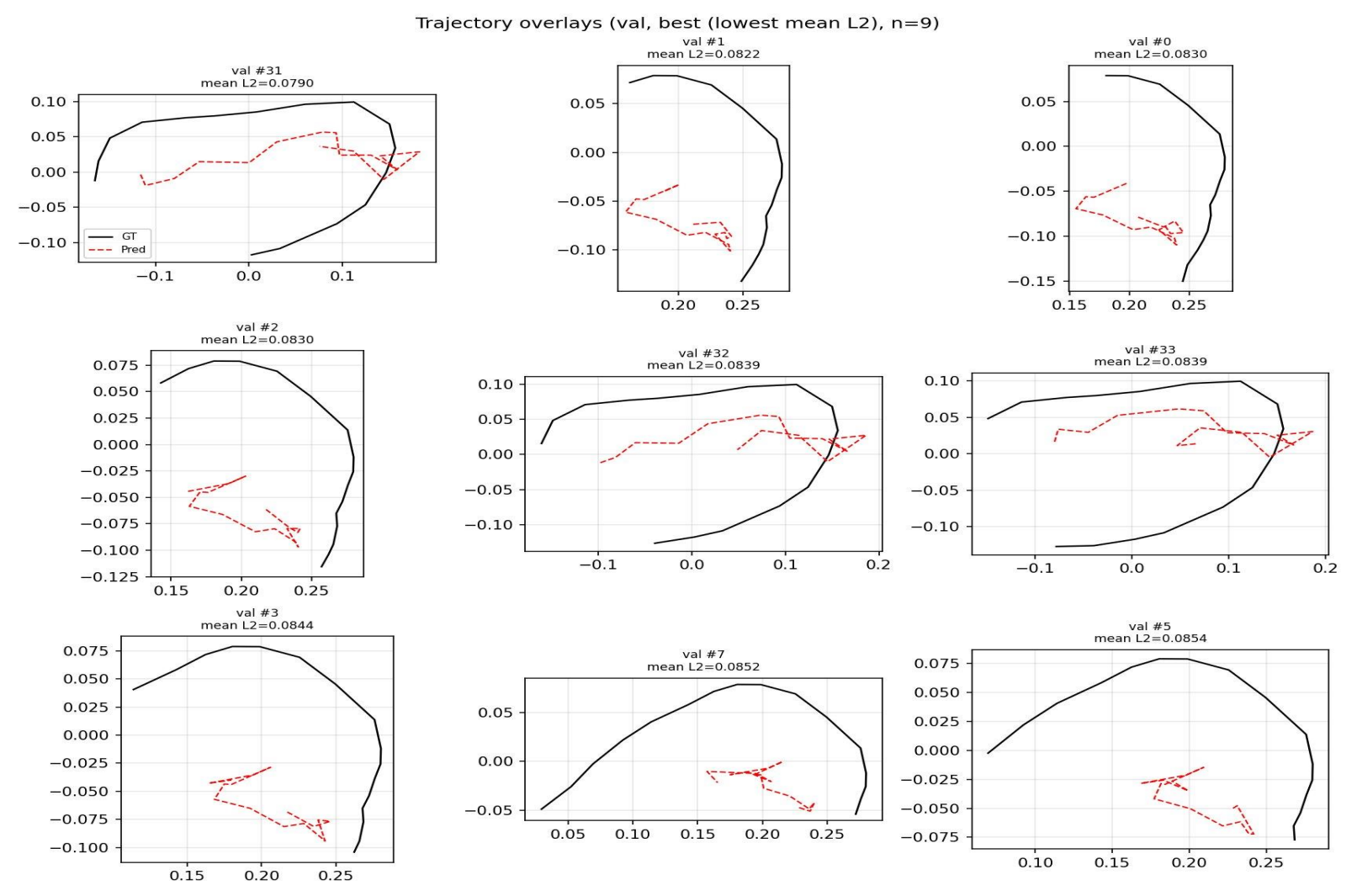
Why we report temporal-split

split	val loss	verdict
merged_random	0.00039	*leaks*
temporal	**0.131 RMSE**	**honest**

Random-window splits place sample k in train and k+1 in val — and consecutive windows share 15 of 16 timesteps. Temporal split holds out a full tail trajectory and is the only fair measure of generalization.

Scale check — WRIST(x, y) lives in a normalized tabletop frame with usable range ≈ ±0.25 along x and ±0.15 along y. An L2 of 0.131 is therefore **~26 % of the half-width** — the model recovers rough location, not precise tracking. The 2.68 % hit-rate at ε = 0.02 shows fine-grained localization has not emerged yet.

GT vs predicted trajectory — best 9 val windows

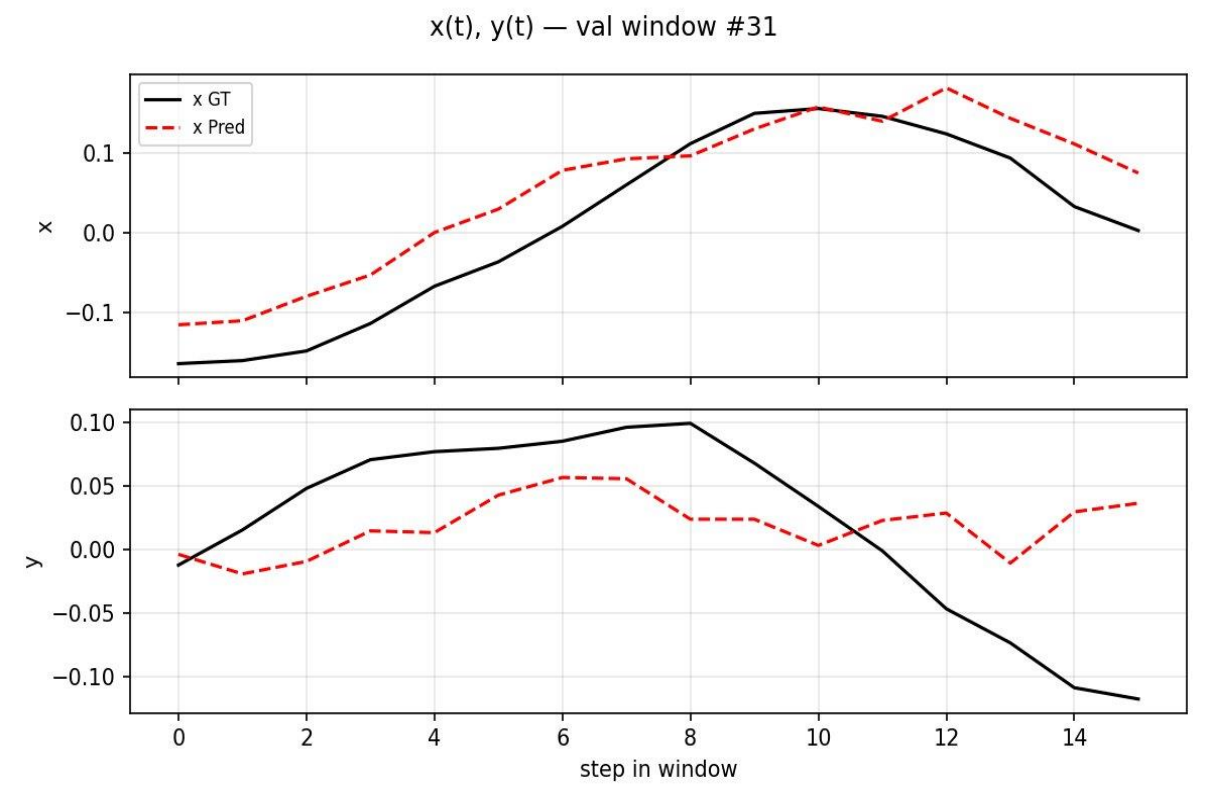


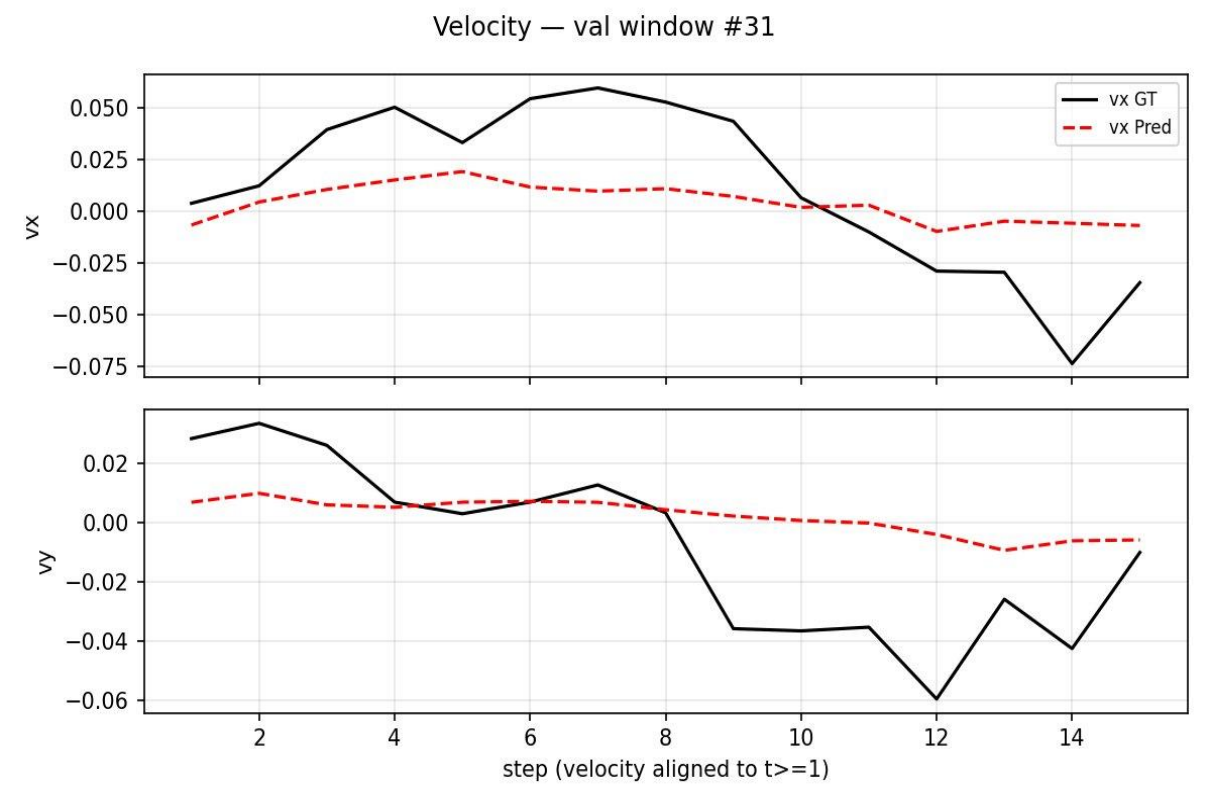
What to see

All panels show temporal-split validation — the **hard** test.

- Black — ArUco ground truth.
- Red dashed — model prediction.
- Prediction clusters near the workspace centroid; GT traces much larger loops (RMSE 0.131).
- Motion shape is partially recovered (see val #31) but amplitude is heavily shrunk.
- *A classic regress-to-the-mean signature driven by a tiny dataset (435 samples, 42 val windows).*

x(t), y(t) and velocity on val window #31





Position tracks the trend but flattens

x follows the rising-then-falling GT, y loses the second descent. The prediction is biased toward the workspace centroid.

Velocity head under-estimates magnitude

Predicted v_x, v_y oscillate tightly around 0 while GT reaches ±0.07. The tanh·vel_scale = 0.08 bound is not the cause — the model simply has not learned the mapping.

What is limiting the model right now?

1. Dataset is tiny

Two sessions, ~435 samples, ≈84 valid windows after split — far below what a transformer + GRU with ~100k params needs to generalize. The model memorizes the mean hand location.

2. Calibration gap

Phase per tag/antenna pair has an unknown integer offset. The 2048-period wrap is modeled, but per-channel bias is left for the network to discover from 435 points.

3. Velocity signal is weak

Because position loss dominates early training, $\mathcal{L}$_vel gradient is crowded out. vel-head stays near the init and the dynamics branch becomes a trivial 'carry-over' of the previous position.

4. Split-policy amplifies the issue

With only ~210 windows, every partition is fragile. A random split gives an artificially rosy 0.00039 val loss by leaking adjacent frames; temporal split exposes the true 0.131 RMSE. Either way, 42 val windows is too few to steer the model.

Three paths to push L2 below 0.03

01

Collect 10× more data

Run 20+ sessions across different users and hand speeds. 435 samples is a prototype — real training needs thousands.

02

Per-channel phase calibration

Measure the static phase offset per tag/antenna pair with the hand removed, subtract it before feeding the network, and let the model learn only motion-induced phase.

03

Physics-anchored decoder

Replace the free MLP position head with a triangulation-style layer that reads tag positions and constrains $\hat{p}$ to lie on the surface — narrows the solution space.

Additional levers: multi-loss rebalancing (schedule $\mathcal{L}$_vel ↑ after $\mathcal{L}$_pos plateaus); data augmentation on the phase channel; knowledge distillation from an EKF/particle-filter baseline.

Takeaways

A dual-branch tracker is the right scaffold. The bottleneck is data, not architecture.

Built end-to-end

From raw CSV through 14-feature channels to a 2-branch model trained with 4 losses.

Shape, not amplitude

The model captures the curvature of hand motion but shrinks toward the workspace center.

Next step is data

10× data + per-channel phase calibration should cut mean-L2 by 2-3× with no model changes.

Thank you · Questions?